[TEAM NAME]

AIAA DESIGN/BUILD/FLY 2026-27 / PROPOSAL

Towed-Sensor Surveillance UAV

AMERICAN UNIVERSITY OF THE MIDDLE EAST

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1.0 EXECUTIVE SUMMARY

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DESIGN AT A GLANCE

CONCEPT RENDER

WINGSPAN	M2 GROSS WEIGHT
`[span]` ft	`[TOGW M2]` lb
SENSOR WEIGHT	M2 PAYLOAD
`[W sensor]` lb	`[units]` units
M2 FIVE-LAP TIME	M3 LAPS
`[5-lap time]` s	`[laps]` laps
GM DROP HEIGHT	PRED. MISSION SCORE
`[h]` in	`[total]` / 7.50

2.0 MANAGEMENT SUMMARY

2.1 Team Organization

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2.2 Schedule and Milestones

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TABLE 01 Sub-team responsibilities and the skill sets each requires.

SUB-TEAM	RESPONSIBILITY	REQUIRED SKILLS
Aero & Performance	Sizing, lap and energy model, stability and control	[skills]
Structures & Manufacturing	Airframe design, layup and build, load tests	[skills]
Propulsion & Avionics	Motor, propeller, battery, ESC, radio and fail-safe	[skills]
Sensor Systems	Sensor, D&R reel, shipping container, rigging	[skills]
Flight Operations	Pilot, observer, ground crew, checklists	[skills]
Business & Logistics	Budget, sponsorship, travel, documentation	[skills]

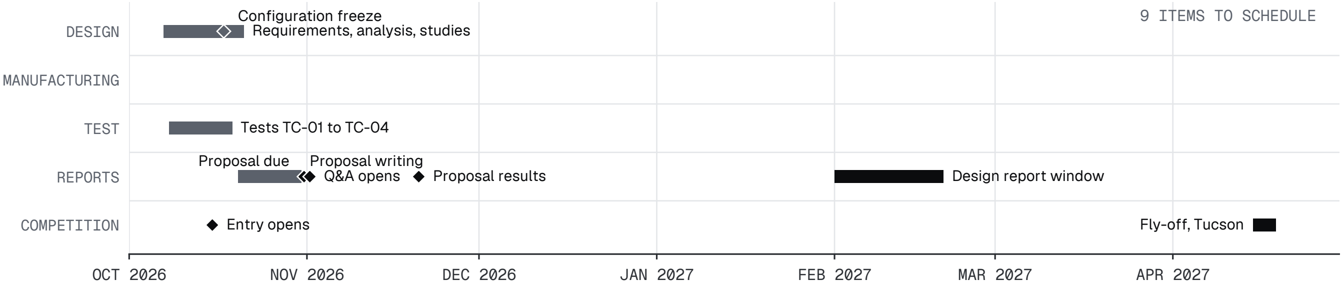

FIG. 01 [takeaway]. Milestones: proposal 31 Oct; Q&A rounds from 1 Nov; PDR; CDR; prototype first flight; sensor bench test; design report 19 Feb; competition aircraft first flight; Pre-Tech/First-Flight certification; fly-off 15–18 Apr.

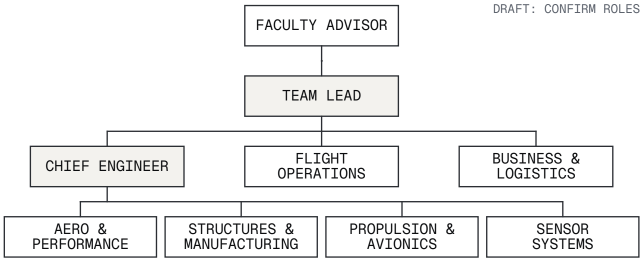

FIG. 02 [takeaway]: one owner for every scored subsystem.

TABLE 02 Program budget, all costs including travel (USD).

CATEGORY	COST	SHARE
Airframe, prototypes, test rigs	$1,800	[%]
Propulsion, avionics, radio	$1,914	[%]
Batteries, charging sacks, fuses	$1,029	[%]
Sensor electronics and lights	$547	[%]
Containers, simulators, drop rig	$245	[%]
Spares, props, adhesives, consumables	$357	[%]
Crating materials	$150	[%]
Travel and aircraft freight	[$]	[%]
Fly-off fee, AIAA/AMA memberships	[$]	[%]
Contingency ([%])	[$]	[%]
Total	[$]	**100%**

2.3 Budget

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TABLE 03 Funding sources and status.

SOURCE	AMOUNT	STATUS
[college allocation]	[$]	[confirmed?]
[sponsor]	[$]	[pending?]
Total	[$]	

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3.0 CONCEPTUAL DESIGN APPROACH

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3.1 Mission Requirements Decomposition

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`TABLE 04` Mission requirements decomposed into subsystem requirements. Requirement IDs (SR) are cited in later trades and design decisions. `[confirm SR list]`

MISSION	SCORE	KEY CONSTRAINTS	DERIVED SUBSYSTEM REQUIREMENTS
M1 Staging	$M_1 = 1.0$	3 laps in 5 min; no payload; may be waived and credited by a successful M2	SR-01 stable and controllable empty; SR-02 full assembly and checks in the 5-min staging box
M2 Delivery	$M_2 = 1 + \frac{W/t}{(W/t)_{\max}}$	Sensor in container plus simulators; 5 laps timed; TOP face up; no friction-only retention	SR-03 bay sized for N units; SR-04 positive mechanical restraint; SR-05 CG in limits for every load case
M3 Sensor	$M_3 = 2 + \frac{NW_s}{(NW_s)_{\max}}$	Deploy before first downwind turn; lights OFF, SOLID, FLASHING, OFF; stow before landing	SR-06 internal powered reel, tow $\geq 1.5b$; SR-07 light command over the tow line; SR-08 sensor aerodynamically stable
Ground	$GM = 0.5 + \frac{W_s h}{(W_s h)_{\max}}$	6 drops onto concrete, $h \leq 60$ in; rigging changes within 5 min; zero sensor damage	SR-09 container absorbs $W_s h$; SR-10 quick-swap O/E rigging, flush attachment
System	–	Span ≤ 6 ft; TOGW < 55 lb; ≤ 100 Wh; no autopilot, GPS or camera; positive hatch latches	SR-11 pilot-in-the-loop only; SR-12 latches without magnets; SR-13 energy for M2 at full payload

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3.2 Scoring Model and Sensitivity Study

The total mission score is the sum of four mission scores, each a fixed completion credit plus a term normalized by the best team in the field:[1]

$$S_{\mathrm{M}} = M_1 + \left[1 + \frac{(W/t)}{(W/t)_{\max}}\right] + \left[2 + \frac{NW_s}{(NW_s)_{\max}}\right] + \left[0.5 + \frac{W_s h}{(W_s h)_{\max}}\right] \tag{1}$$

SENSITIVITY / SPIDER OR TORNADO / COLUMN

`FIG. 03` `[takeaway]`: which parameter moves total mission score most.

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3.3 Configuration Trade Studies

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[1] AIAA, *2026–27 Design, Build, Fly Rules*, Draft, Sep. 2026, Sec. 3.3.

CARPET: SCORE VS SENSOR WEIGHT AND UNITS

FIG. 04 [takeaway]: the chosen design point on the coupled M2/M3/GM trade.

TABLE 05 Configuration decisions and the requirement that drove each.

DECISION	OPTIONS CONSIDERED	SELECTED	DRIVER
Wing position	Low, mid, high	[sel.]	[SR]
Tail	Conventional, T, V	[sel.]	[SR]
Landing gear	Tricycle, taildragger	[sel.]	[SR]
Propulsion	Tractor, pusher, twin	[sel.]	[SR]
Sensor bay	Door, open slot	[sel.]	SR-06

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3.4 Preliminary Design and Sizing

3.4.1 Sizing method Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout.

TABLE 06 Deployment-and-recovery architecture trade (scores 1–5; selected option inverted).

CRITERION	WT.	OPT. A	OPT. B	OPT. C
Recovery reliability	[w]	–	–	–
Signal path integrity	[w]	–	–	–
Deploy/stow time	[w]	–	–	–
Weight and volume	[w]	–	–	–
Manufacturability	[w]	–	–	–
Weighted total	**1.00**	**–**	**–**	**–**

CONSTRAINT DIAGRAM / DESIGN POINT

FIG. 05 [takeaway]: which requirement sets T/W and W/S.

3.4.2 Aircraft concept Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout.

CONCEPT 3-VIEW / SENSOR BAY VISIBLE

FIG. 06 [takeaway]: the configuration and where the sensor exits.

TABLE 07 Preliminary design parameters.

PARAMETER	VALUE	UNIT
Wingspan	[span]	ft
Wing area	[S]	ft²
Aspect ratio	[AR]	–
Empty weight	[W empty]	lb
Gross weight, M2 / M3	[TOGW M2] / [TOGW M3]	lb
Battery energy	[Wh]	Wh
Cruise speed	[V cruise]	ft/s
Sensor weight (max)	[W sensor]	lb
Tow-line length	[tow length]	ft

TABLE 08 Mission targets and predicted scores.

MISSION	TARGET	SCORE
M1 Staging	[waive or fly]	1.00
M2 Delivery	[units] units, [W payload] lb, [5-lap time] s	[M2]
M3 Sensor	[laps] laps at [W sensor] lb	[M3]
GM Ground	[h] in at [W sensor] lb	[GM]
Total mission score		[total]

3.4.3 **Sensor and D&R system** Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. The aircraft is flown pilot-in-the-loop at all times: no autopilot or flight controller is carried, and stabilization is limited to optional rate gyros.

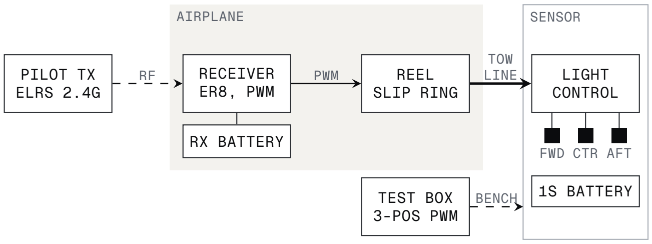

FIG. 07 [takeaway]: the command path to the sensor lights.

3.4.4 **Shipping container and rigging** Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. Placeholder copy sized to this section's word budget for layout. All hatches and the container lid use positive mechanical latches; magnets are never the latching method.

3.5 Mission Design Approach and Scoring Targets

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4.0 MANUFACTURING PLAN

4.1 Preliminary Manufacturing Flow

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MANUFACTURING FLOW / 3 LANES X 4 STAGES

FIG. 08 [takeaway]: parallel airframe, sensor and container lanes converge at integration.

TABLE 09 Critical processes, why they matter, and risk mitigation.

PROCESS	WHY CRITICAL	MITIGATION
Reel slip ring	Signal and tow on one line	Bench cycling on a spare receiver
Sensor body	Stability, drop survival	Blower-rig screen (TC-01)
Container liner	Absorbs $W_s h$	Drop series to 60 in (TC-02)
Wing spar	Tip-load test at MGTOW	Prototype tip-lift proof test

4.2 Critical Processes and Technologies

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5.0 TEST PLANNING

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TABLE 10 Component, ground and flight test plan.

TEST	PASS CRITERION	DATE
Ground and component		
Thrust stand	Amps $<$ fuse rating	[date]
Wing-tip lift at MGTOW	No damage (§4.2.4)	[date]
Light visibility in sun	$\geq 90\,\%$ at 600 ft	[date]
Tow-line signal	3 of 3 modes	[date]
D&R cycle	$\geq 1.5b$ out, stows	[date]
Drop series to 60 in	No damage, 3/3 lit	[date]
GM drill, 5 min	Worst run $\leq$ 4:00	[date]
Staging drill, 5 min	Worst run $\leq$ 4:00	[date]
Range, fail-safe, CG	§4.2.5 checks pass	[date]
Flight		
Handling and trim	Level-flight video	[date]
Sensor tow stability	No rotation	[date]
Stepped M2 payload	5 laps $<$ 5 min	[date]
Full M3 sequence	§3.3.3.3 sequence	[date]
Energy per mission	$\geq 20\,\%$ reserve	[date]